

Trading Energy Futures with Inflation (Working Draft)

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1 Introduction

This repo¹ and research began with the idea of linking inflation to commodities, specifically energy. It is well known that energy futures and inflation are closely related, yet there is relatively little research modeling inflation as a factor within commodity markets, specifically futures. Existing research on the relationship between energy and inflation primarily focuses on energy shocks and the relationship between CPI and spot prices, often emphasizing broader macroeconomic implications. This research instead examines inflation using a higher-frequency measure, with a particular focus on its relationship with commodity futures and the development of a strategy.

The original motivation for this paper and strategy originate from a specific graph in a Federal Reserve Bank of St. Louis article by Andolfatto and Ebsim [1]. The article examined the relationship between spot crude oil prices and the U.S. 5-year, 5-year forward inflation expectation rate. The strength of the relationship suggested that it could potentially be extended into a tradable model. This is of interest since 5-year, 5-year forward inflation expectations are available daily and therefore at a higher frequency than CPI, and crude oil spot prices.

2 Data

The original research proposed by Andolfatto and Ebsim[1] used WTI spot prices which are included in this analysis. Other relevant energy spot prices and their futures are included to ensure consistency and to examine whether the relationship extends beyond WTI. The commodity futures considered are liquid and actively traded crude oils and refined energies contracts, specifically WTI, Brent, Natural Gas, Heating Oil, RBOB, and Gas Oil. These contracts were selected since they cover the most commonly traded futures in the energy market.

The data comes from several sources. The first source is the Federal Reserve Bank of St. Louis (FRED), which is used for consistency when verifying the original results from the article. The 5y5y forward inflation and energy spot prices are collected from FRED.

¹<https://github.com/diegodalvarez/CommodityForwardInflation>

Bloomberg Terminal is used for the futures data and the other inflation measurements. The 5y5y forward rate is collected (again) along with Bloomberg Economic Inflation Surprise indexes. The energy futures prices are collected from Bloomberg, which are all front month roll adjusted via ratio through Bloomberg’s roll adjustment methodology.

The following data is collected from St. Louis Federal Reserve

Name	Start Date	End Date
Brent	1989-01-03	2026-07-27
WTI	1999-01-04	2026-07-27
Natural Gas	1997-01-07	2026-07-27
Heating Oil	1999-01-04	2026-07-27
RBOB	2003-03-11	2026-07-27
5y5y Forward Inflation	2003-01-02	2026-08-03

Table 1: FRED Data

For the front month futures Data from Bloomberg

	Start Date	End Date
security		
CL1	1990-01-02	2024-10-29
CO1	1990-01-02	2024-10-29
HO1	1990-01-02	2024-10-29
NG1	1990-04-04	2024-10-29
QS1	1990-01-02	2024-10-29
XB1	2005-10-04	2024-10-29

Table 2: FRED Data

For the inflation-related measurements

	Start Date	End Date
security		
GBP Inflation Swap Forward 5Y5Y	2004-04-27	2024-10-28
UK Bloomberg Economics Inflation Data Surprise	2001-01-01	2024-10-28
US Bloomberg Economics Inflation Data Surprise	2000-01-03	2024-10-28
USD Inflation Swap Forward 5Y5Y	2004-07-21	2024-10-28

Table 3: Inflation Measurements

Some of the datasets have different windows. This was from different parts of the repo being written (and programmed) at different times. Ultimately this will have to be fixed, but for the time being the results should be relatively consistent without the missing ~ 2 years of data.

3 Background Analysis

First start by examining the data from the Federal Reserve of St. Louis article. The authors model the relationship in the opposite direction comparing the raw spot prices with forward inflation. They only use WTI and 5y5y forward, but it can be extended to other spot energy prices. The graph from the article is shown below.

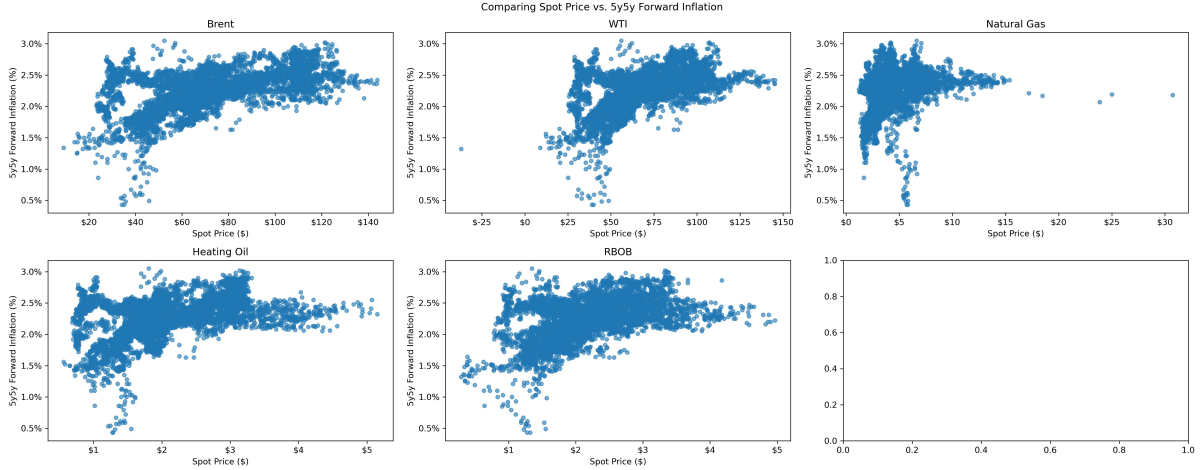


Figure 1: recreating of FRED article graph comparing 5y5y forward inflation to spot energy prices (FRED Dataset)

This relationship between level price and rate appears relatively stable, as both time series exhibit some degree of boundedness². However, the relationship should be examined through a regression using stationary variables. For spot prices, daily changes are calculated and normalized by their 10-day exponentially weighted standard deviation. For the 5y5y forward inflation the daily logged differences are calculated.

Separate regressions are run to compare instantaneous and lagged 5y5y forward inflation changes. The table below shows the statistics regarding the β of the OLS. The β is statistically significant for the instantaneous scenario but that relationship does not hold when it is lagged.

Lag Energy	Coeff.		p -value		t -value	
	No Lag	Lagged	No Lag	Lagged	No Lag	Lagged
Brent	3.68	0.70	0.00	0.11	8.06	1.59
Heating Oil	3.26	0.36	0.00	0.39	7.18	0.87
Natural Gas	0.10	-0.24	0.82	0.55	0.22	-0.59
RBOB	2.22	0.45	0.00	0.39	4.95	0.86
WTI	3.40	0.83	0.00	0.09	7.53	1.68

Table 4: β value of OLS regression of log differences in 5y5y forward inflation to risk normalized differences in spot prices

²Notwithstanding periods of hyperinflation

Logistic regression was included to get better p-values with lagged forward inflation, which ultimately didn't fare better than the OLS model.

Lag Energy	Coeff.		<i>p</i> -value		<i>z</i> -value	
	No Lag	Lagged	No Lag	Lagged	No Lag	Lagged
Brent	9.36	1.48	0.00	0.16	6.83	1.41
Heating Oil	7.69	0.48	0.00	0.62	5.82	0.49
Natural Gas	1.68	-1.11	0.12	0.27	1.57	-1.10
RBOB	4.85	0.91	0.00	0.36	3.86	0.91
WTI	8.51	2.13	0.00	0.05	6.35	1.94

Table 5: β value of Logistic regression of log differences in 5y5y forward inflation to risk normalized differences in spot prices

When measuring the relationship between 5y5y forward inflation and energy futures many articles tend to use price and yield level for rolling correlations. Below is a graph of the rolling 1y correlation between price level and 5y5y forward inflation.

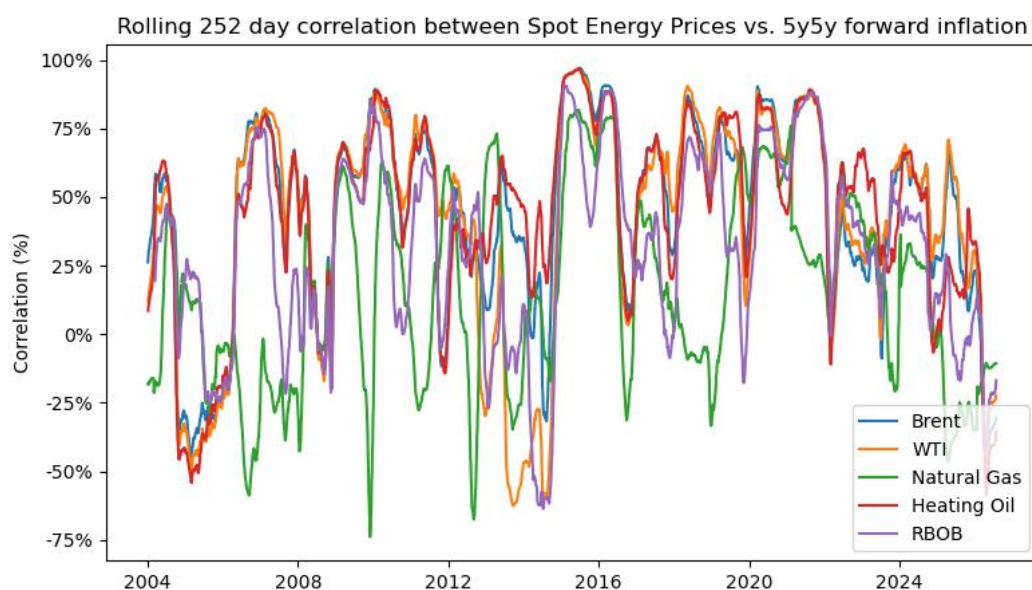


Figure 2: 1y rolling correlation between spot prices and 5y5y forward inflation (FRED Dataset)

The correlation can vary and at times can be negative which has led some to believe that the correlations are not stable. When differencing the values and then running a rolling correlation shows that it's usually positive.

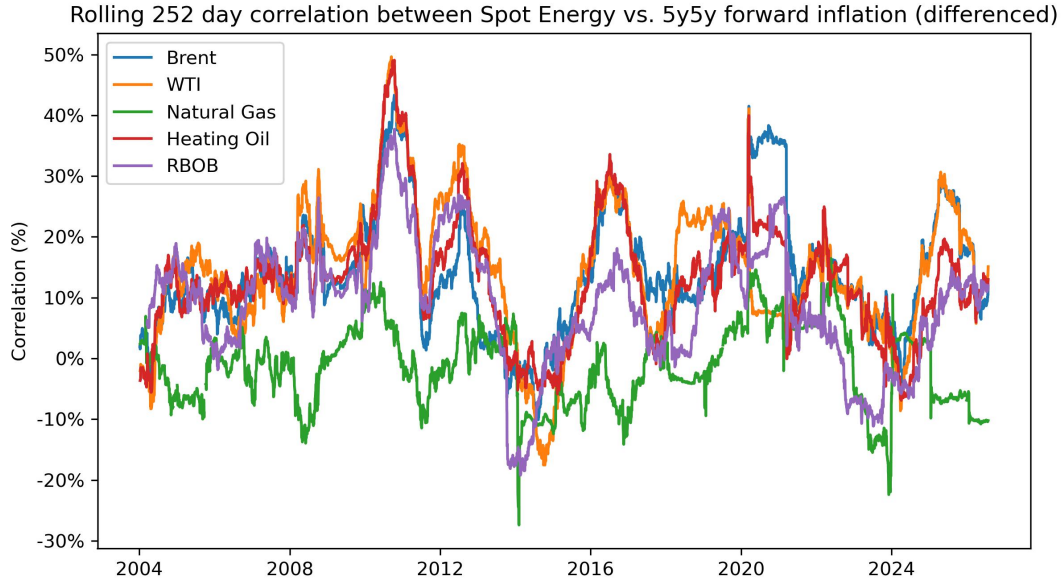


Figure 3: 1y rolling correlation between differenced spot prices and 5y5y forward inflation (FRED Dataset)

Below is a table of some of the aggregated states for the rolling 1y correlations. They show that on average the correlations are positive, although a bit weak. For the historical time series, the data is almost always positive covering $\sim 90\%$ of the whole dataset.

	Lowest	Highest	Average	Median	% > 0
Brent	-0.10	0.43	0.14	0.13	0.94
WTI	-0.18	0.50	0.14	0.15	0.91
Natural Gas	-0.27	0.16	-0.01	-0.01	0.45
Heating Oil	-0.07	0.49	0.13	0.13	0.90
RBOB	-0.19	0.38	0.09	0.10	0.85

Table 6: Aggregate statistics of rolling 1y correlations of log differences of 5y5y forward vs spot differences

The next step is to extend this to the tradable dataset. Start by running the same analysis but using the futures contracts as opposed to the spot prices collected from Fred. When analyzing results with futures, especially commodities, volatility targeting is used. Controls for contemporaneous and lagged hedging are taken into consideration. All futures are targeted at a 10% annualized volatility using 100 day exponentially weighted standard deviation.

The table below shows statistics regarding the β of the OLS regression for the commodity futures. This regression is run instantaneously therefore it is measuring the pure factor exposure of the inflation signal. All regressions are ran separately.

Commod	Inf Measure Vol Target	Coeff		<i>p</i> -value		<i>t</i> -value	
		UK 5Y5Y	US 5Y5Y	UK 5Y5Y	US 5Y5Y	UK 5Y5Y	US 5Y5Y
CL1	Lag	0.01	0.01	0.00	0.00	4.15	5.35
	Perf	0.01	0.01	0.00	0.00	4.17	5.23
CO1	Lag	0.01	0.01	0.00	0.00	4.38	5.17
	Perf	0.01	0.01	0.00	0.00	4.41	5.04
HO1	Lag	0.01	0.01	0.00	0.00	4.27	3.79
	Perf	0.01	0.01	0.00	0.00	4.63	4.09
NG1	Lag	0.00	0.00	0.56	0.31	0.58	1.01
	Perf	0.00	0.00	0.37	0.12	0.90	1.54
QS1	Lag	0.01	0.01	0.00	0.00	5.74	5.52
	Perf	0.01	0.01	0.00	0.00	6.09	5.91
XB1	Lag	0.01	0.01	0.00	0.00	4.91	4.10
	Perf	0.01	0.01	0.00	0.00	5.13	4.26

Table 7: OLS β for 5y5y forward inflation and commodity futures (vol-hedged) (Bloomberg)

Below is the corresponding table for the Bloomberg Economics measurement using the same approach. As a note the Bloomberg Economics dataset was included for its trading ability and thus there is likely less relationship between returns and the index, which is shown in the table below.

Commod	Inf Measure Vol Target	Coeff		<i>p</i> -value		<i>t</i> -value	
		UK Sur	US Sur	UK Sur	US Sur	UK Sur	US Sur
CL1	Lag	-0.00	-0.00	0.78	0.97	-0.28	-0.04
	Perf	-0.00	-0.00	0.76	0.99	-0.31	-0.01
CO1	Lag	-0.00	0.00	0.54	0.71	-0.61	0.37
	Perf	-0.00	0.00	0.52	0.68	-0.64	0.41
HO1	Lag	-0.00	-0.00	0.58	0.78	-0.56	-0.28
	Perf	-0.00	-0.00	0.55	0.80	-0.60	-0.25
NG1	Lag	0.00	-0.00	0.98	0.82	0.03	-0.23
	Perf	0.00	-0.00	0.90	0.48	0.13	-0.70
QS1	Lag	0.00	0.00	0.79	0.83	0.27	0.21
	Perf	0.00	0.00	0.79	0.80	0.27	0.25
XB1	Lag	-0.00	0.00	0.55	0.59	-0.59	0.54
	Perf	-0.00	0.00	0.54	0.57	-0.62	0.57

Table 8: OLS β for Bloomberg Economics Inflation Surprise and commodity futures (vol-hedged) (Bloomberg)

Now that that instantaneous relationship has been analyzed the next step is lag signal re-run the regression and compare the results³.

³Full results of the lagged regression are available in the repo

Inf Ticker	Lag Commod	Coeff		<i>p</i> -value		<i>t</i> -value	
		Perf	Lag	Perf	Lag	Perf	Lag
UK 5Y5Y	CL1	0.01	0.00	0.00	0.21	4.17	1.26
	CO1	0.01	0.00	0.00	0.20	4.41	1.29
	HO1	0.01	0.00	0.00	0.11	4.63	1.59
	NG1	0.00	0.00	0.37	0.80	0.90	0.26
	QS1	0.01	0.00	0.00	0.06	6.09	1.89
	XB1	0.01	0.00	0.00	0.52	5.13	0.65
UK Surprise	CL1	-0.00	-0.00	0.76	0.13	-0.31	-1.50
	CO1	-0.00	-0.00	0.52	0.05	-0.64	-1.95
	HO1	-0.00	-0.00	0.55	0.02	-0.60	-2.32
	NG1	0.00	-0.00	0.90	0.69	0.13	-0.41
	QS1	0.00	-0.00	0.79	0.01	0.27	-2.57
	XB1	-0.00	-0.00	0.54	0.12	-0.62	-1.56
US 5Y5Y	CL1	0.01	0.00	0.00	0.18	5.23	1.33
	CO1	0.01	0.00	0.00	0.08	5.04	1.74
	HO1	0.01	0.00	0.00	0.13	4.09	1.52
	NG1	0.00	0.00	0.12	0.39	1.54	0.86
	QS1	0.01	0.00	0.00	0.05	5.91	1.93
	XB1	0.01	0.00	0.00	0.17	4.26	1.38
US Surprise	CL1	-0.00	-0.00	0.99	0.22	-0.01	-1.23
	CO1	0.00	-0.00	0.68	0.29	0.41	-1.06
	HO1	-0.00	-0.00	0.80	0.20	-0.25	-1.29
	NG1	-0.00	-0.00	0.48	0.33	-0.70	-0.98
	QS1	0.00	-0.00	0.80	0.32	0.25	-0.99
	XB1	0.00	-0.00	0.57	0.76	0.57	-0.31

Table 9: OLS β comparing Bloomberg Economics Inflation Surprise and 5y5y against commodity futures (vol-hedged, and contemporaneously) controlling for lagged exogenous factors (Bloomberg)

It is evident that the β values are quite small even though most of them are statistically significant within OLS. The results are relatively consistent with OLS models in daily returns data.

4 Trading Z-Scores

This section involves generating a generic signal and then backtesting it on a single-security basis. The approach It uses the same approach in the prior but uses a 30 day exponentially weighted z-score. First start by calculating the 30-day exponentially weighted z-score of the inflation variables and then applying them to the commodity futures.

Before examining strategy results an initial regression between the z-scored values and the commodity returns is required. Contemporaneously volatility targeted returns are

used⁴. One regression is run instantaneously and the other is ran with the signal lagged to measure its predictive capabilities.

Ticker	Country	Lag Signal	Coeff		<i>p</i> -value		<i>t</i> -value	
			Lag	No Lag	Lag	No Lag	Lag	No Lag
CL1	UK	5y5y	0.00	0.01	0.10	0.00	1.63	5.94
		Sur	-0.00	-0.00	0.34	0.90	-0.96	-0.13
	US	5y5y	0.00	0.01	0.00	0.00	2.98	5.93
		Sur	-0.00	-0.00	0.10	0.18	-1.62	-1.33
CO1	UK	5y5y	0.00	0.01	0.06	0.00	1.85	5.78
		Sur	-0.00	-0.00	0.15	0.67	-1.45	-0.43
	US	5y5y	0.00	0.01	0.00	0.00	3.30	5.89
		Sur	-0.00	-0.00	0.22	0.38	-1.23	-0.87
HO1	UK	5y5y	0.00	0.01	0.04	0.00	2.10	6.02
		Sur	-0.00	-0.00	0.05	0.39	-1.96	-0.87
	US	5y5y	0.00	0.01	0.00	0.00	3.05	4.59
		Sur	-0.00	-0.00	0.20	0.22	-1.28	-1.22
NG1	UK	5y5y	0.00	0.00	0.76	0.23	0.31	1.21
		Sur	-0.00	-0.00	0.79	0.96	-0.27	-0.06
	US	5y5y	0.00	0.00	0.63	0.12	0.48	1.56
		Sur	-0.00	-0.00	0.07	0.26	-1.78	-1.12
QS1	UK	5y5y	0.00	0.01	0.05	0.00	1.98	7.92
		Sur	-0.00	0.00	0.01	0.40	-2.49	0.84
	US	5y5y	0.01	0.01	0.00	0.00	3.51	5.78
		Sur	-0.00	-0.00	0.22	0.28	-1.24	-1.07
XB1	UK	5y5y	0.00	0.01	0.27	0.00	1.09	6.50
		Sur	-0.00	-0.00	0.13	0.34	-1.51	-0.96
	US	5y5y	0.00	0.01	0.02	0.00	2.32	4.05
		Sur	-0.00	-0.00	0.51	0.77	-0.67	-0.29

Table 10: β value of OLS regression z-scored signal to contemporaneously volatility targeted returns

The results are quite similar to the previous regression results which can be found in table 9. The normalization through z-scoring did make some of exogenous variables statistically significant.

The first backtest will begin with trading the signal solely by itself, without an presumption of being long or short. The sign of the signal is applied to the raw returns (without targeting) and the results signal returns are then volatility targeted. Like all volatility targeting throughout this paper, contemporaneous and lagged hedging are taken into consideration. Below is a table of the sharpes for each security.

⁴see repo for lagged volatility targeting

Vol. Target security	UK 5y5y		UK Sur. Inf.		US 5y5y		US Sur. Inf.	
	Lagged	Perfect	Lagged	Perfect	Lagged	Perfect	Lagged	Perfect
CL1	0.38	0.36	0.06	0.05	0.70	0.72	-0.47	-0.48
CO1	0.47	0.47	0.12	0.11	0.67	0.69	-0.43	-0.44
HO1	0.34	0.43	-0.13	-0.11	0.52	0.64	-0.41	-0.39
NG1	0.37	0.30	-0.22	-0.11	0.25	0.13	0.05	-0.32
QS1	0.44	0.52	0.05	0.02	0.76	0.88	-0.35	-0.33
XB1	0.20	0.27	-0.14	-0.19	0.46	0.55	-0.41	-0.37

Table 11: Sharpe of single-security performance of raw z-scores applied to futures.

Then taking equal weight of the single-security grouping per each signal leads to the results.

Vol. Target	UK 5y5y	UK Sur. Inf.	US 5y5y	US Sur. Inf.
Lagged	0.58	-0.05	0.79	-0.40
Perfect	0.46	0.02	0.77	-0.48

Below is a plot of the returns which shows steady returns for the 5y5y forward strategies. Regarding the Bloomberg Economics Surprise Inflation Index there are weakly negative returns (in terms of absolute sharpe).

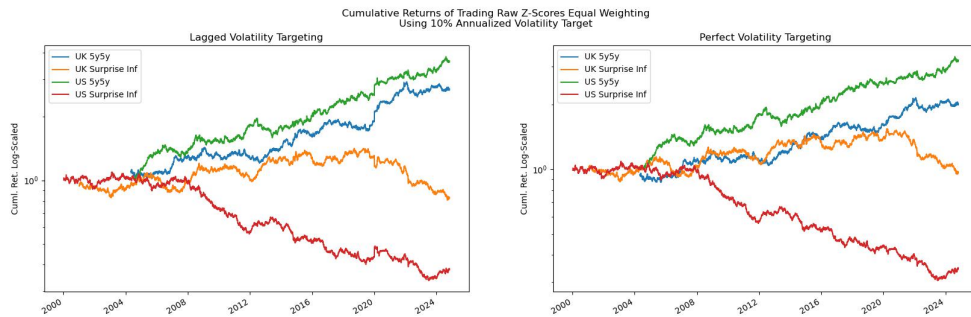


Figure 4: Performance of raw z-scores applied to energy futures, equal weighting, 10% annualized volatility

To ensure the signal is not driven by the trend effect or is related with trend following benchmarks the results are compared to them. Using some sample trend following returns indices and running a regression and measuring correlation between the signal returns and trend shows a weak relationship and sometimes no correlation.

Trend Idx	Name	Corr	Coeff	<i>p</i> -value	<i>t</i> -stat
BBG Commod	5y5y UK	0.03	0.06	0.05	1.97
	5y5y US	0.01	0.03	0.30	1.04
	Sur UK	0.02	-0.05	0.09	-1.71
	Sur US	-0.12	-0.29	0.00	-9.44
DB Cross-Asset	5y5y UK	0.01	0.01	0.56	0.58
	5y5y US	0.00	-0.00	0.98	-0.02
	Sur UK	0.04	-0.05	0.01	-2.74
	Sur US	-0.04	-0.06	0.00	-2.94
SG Commod	5y5y UK	0.02	0.02	0.17	1.36
	5y5y US	0.02	0.01	0.18	1.33
	Sur UK	-0.01	0.01	0.60	0.52
	Sur US	-0.10	-0.08	0.00	-7.77
SG Trend	5y5y UK	0.01	-0.01	0.47	-0.73
	5y5y US	0.00	-0.00	0.91	-0.11
	Sur UK	0.01	-0.01	0.48	-0.71
	Sur US	-0.02	-0.01	0.21	-1.25

Table 12: Correlation and β statistics of z-score signal portfolios and trend following benchmarks

5 Trading via OLS

This section will cover the results of strategies generated by fitting the z-scored inflation values to the commodity futures and then trading the resulting output. Two methods are tested, the first is fitting the instantaneous regression and trading the lagged residuals, and the second is trading the lagged z-scored inflation values and trading the forecasted returns. All the regressions are fitted and calculated using volatility-targeted commodity futures returns, then the resulting signal from the OLS is applied to the raw returns. The signal returns are then targeted at a 10% annualized volatility.

Throughout the results the regressions are run Full Sample/In Sample, Train/Test, and Expanding Out-of-Sample. The Train/Test period isn't determined by some ratio and instead is cutoff in 2018 which covers 75% of the dataset.

5.1 Trading OLS Residuals

The idea behind this strategy is to run the instantaneous regression, calculate the residual and taking an opposing trade. This is akin to using OLS to calculate the fair value and then taking a position that the returns will regress back to their expected value. Although the regressions from tables 7 and 8 showed that the signals were statistically significant this strategy failed to generate reasonable returns.

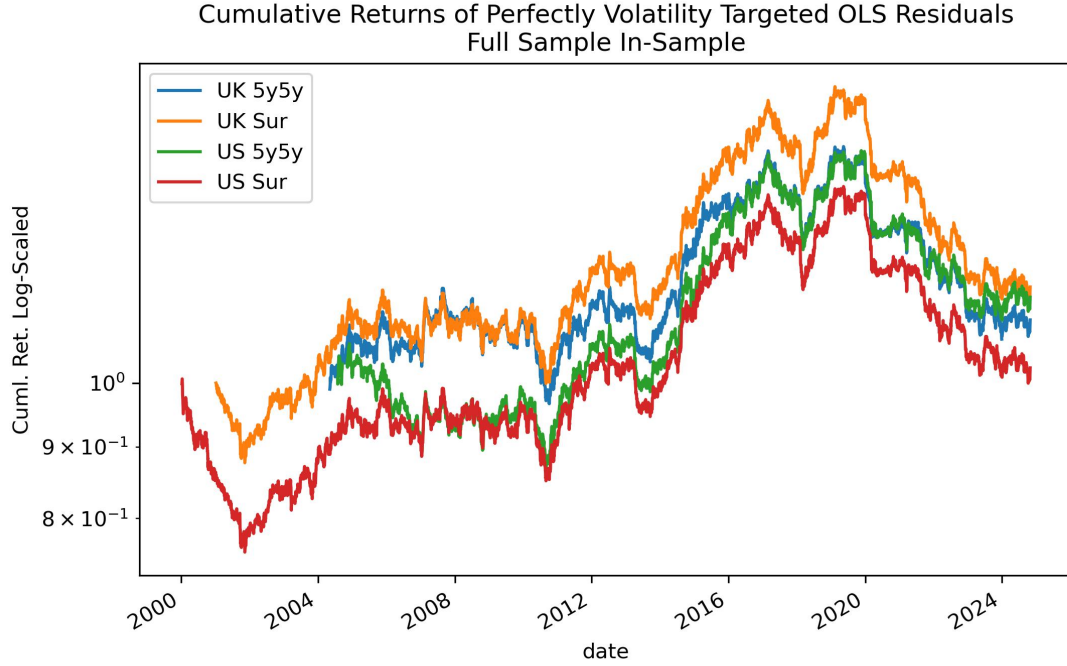


Figure 5: Performance of trading OLS residuals, equal weighting, 10% annualized volatility (hedged contemporaneously)

The plot above is shown for the full sample and perfectly volatility targeted portfolio. Below is the table of sharpes for each time series.

	5y5y	Sur
country		
UK	0.11	0.13
US	0.14	0.05

Table 13: Sharpes of trading OLS residuals, equal weighting, 10% annualized volatility (hedged contemporaneously)

Given the results of the full-sample/in-sample returns were so weak out-of-sample analysis won't be reported⁵

5.2 Trading Forecasted OLS Returns

This approach is very similar to the prior one, except lag the z-score calculating the forecasted returns and then taking a position corresponding to the sign. The performance of this signal is much stronger than the prior. The natural explanation is that the raw z-score approach in section⁶ 4 show that the returns are quite reasonable without any model needed. Therefore, the OLS is working as a mechanism to properly scale the z-score. The z-score approach did not make any assumption about the relationship,

⁵GitHub repo contains the necessary information for the out-of-sample and expanding results

⁶table 11, table 12, and Figure 4

although one could've been implied by the prior regressions, but the results of the OLS forecasted values show that the OLS is properly identifying the relationship between the z-score and returns.

Fut	Signal	Vol. Target Sample Regression Country	Out-Sample Expanding	Lagged		Out-Sample Train/Test	Out-Sample Expanding	Perfect		Out-Sample Train/Test
				In-Sample Full Sample	In-Sample Train/Test			In-Sample Full Sample	In-Sample Train/Test	
CL1	5y5y	UK	0.57	0.44	0.49	0.17	0.58	0.41	0.45	0.11
		US	0.71	0.60	0.44	0.39	0.73	0.60	0.44	0.36
	Sur	UK	0.36	0.23	0.00	0.37	0.40	0.28	0.02	0.48
		US	0.16	0.13	0.06	0.32	0.21	0.19	0.10	0.43
CO1	5y5y	UK	0.49	0.38	0.70	0.11	0.52	0.39	0.71	0.07
		US	0.85	0.80	0.73	0.59	0.86	0.81	0.76	0.58
	Sur	UK	0.41	0.36	0.25	0.66	0.46	0.41	0.26	0.80
		US	0.35	0.29	0.25	0.32	0.41	0.34	0.27	0.44
HO1	5y5y	UK	0.48	0.45	0.76	0.04	0.52	0.43	0.76	0.13
		US	0.73	0.77	0.67	0.54	0.72	0.75	0.68	0.44
	Sur	UK	0.46	0.38	0.19	0.97	0.46	0.39	0.21	0.95
		US	0.37	0.32	0.23	0.80	0.38	0.33	0.22	0.78
NG1	5y5y	UK	0.38	0.26	0.37	-0.07	0.43	0.30	0.44	-0.08
		US	0.44	0.32	0.47	-0.08	0.48	0.37	0.54	-0.09
	Sur	UK	0.38	0.20	0.30	-0.29	0.40	0.23	0.35	-0.31
		US	0.29	0.22	0.45	-0.01	0.31	0.25	0.49	-0.00
QS1	5y5y	UK	0.59	0.48	0.80	0.24	0.63	0.47	0.81	0.34
		US	0.88	0.97	0.90	0.83	0.88	0.96	0.93	0.73
	Sur	UK	0.56	0.50	0.39	0.79	0.57	0.48	0.37	0.76
		US	0.40	0.33	0.28	0.62	0.42	0.34	0.30	0.58
XB1	5y5y	UK	0.38	0.30	0.36	0.57	0.39	0.34	0.37	0.53
		US	0.56	0.55	0.42	0.65	0.56	0.54	0.43	0.60
	Sur	UK	0.46	0.38	0.15	0.91	0.49	0.40	0.16	0.94
		US	0.44	0.24	0.27	0.28	0.46	0.26	0.28	0.29

Table 14: Performance of trading OLS residuals, equal weighting, 10% annualized volatility (hedged contemporaneously)

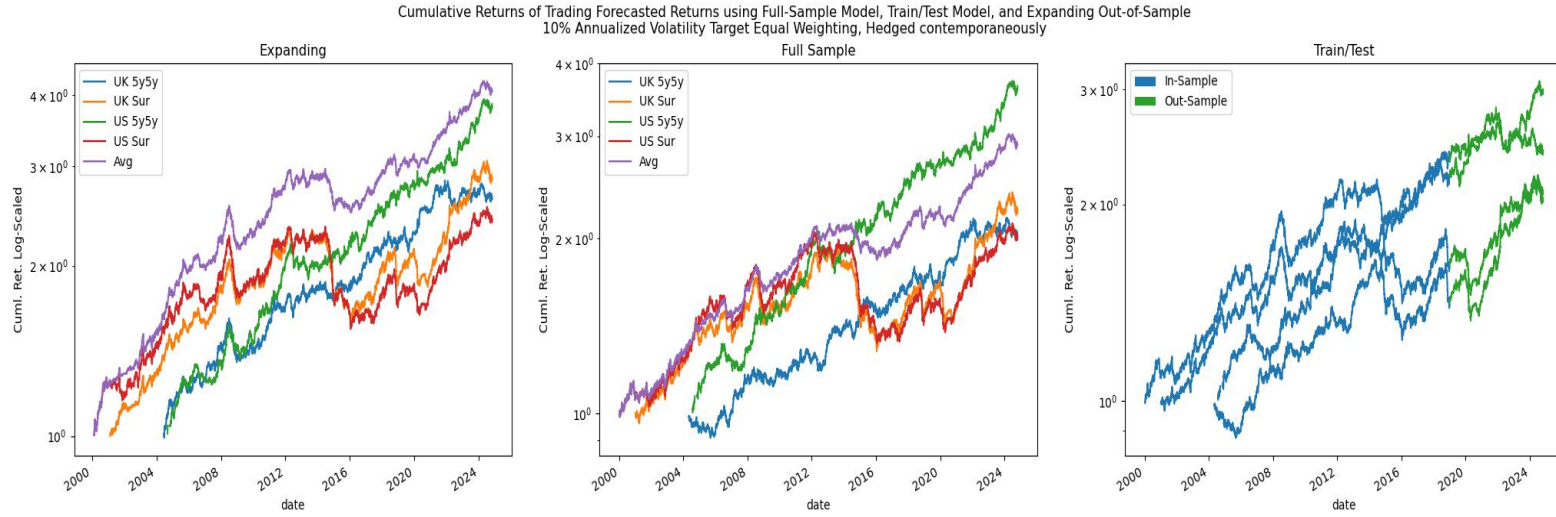


Figure 6: Performance of trading OLS Forecasted values, equal weighting, 10% annualized volatility (hedged contemporaneously)

The results of the portfolio are quite strong when they are put together. Applying equal weight⁷ results in the following Sharpes in the table below. The lagged vs hedging is controls for the volatility targeting and not the signal.

Signal	Regression Country	Lagged			Perfect		
		Exp	FS	Tr/Test	Exp	FS	Tr/Test
5y5y	UK	0.70	0.53	0.66	0.74	0.54	0.66
	US	0.99	0.93	0.77	1.01	0.94	0.80
Sur	UK	0.61	0.45	0.41	0.65	0.49	0.44
	US	0.45	0.37	0.40	0.50	0.41	0.44

Table 15: Sharpes for equal weight portfolio for each signal using forecasted OLS values, controlling for volatility targeting and sample size

The next part is to measure the sharpe difference between in-sample and out-of-sample returns. This is done by taking the subtracting the from sample sharpe from the train/test model and the expanding out-of-sample model. The resulting sharpe loss (or gain) is quite small implying that the model fits reasonably well. The table on the next page shows the sharpe lost controlling for volatility targeting. The results are consistent with losses ranging from ~ 0.1 to ~ 0.3

⁷which is comparable to equal volatility since the returns are volatility targeted

Fut	Country	Regression	Train/Test Loss		Expanding Loss	
		Vol. Target Signal	Lagged	Perfect	Lagged	Perfect
CL1	UK	5y5y	-0.33	-0.36	-0.12	-0.13
		Sur	-0.24	-0.25	0.13	0.12
	US	5y5y	-0.31	-0.32	0.04	0.04
		Sur	0.00	0.00	-0.10	-0.10
CO1	UK	5y5y	-0.10	-0.15	0.11	0.09
		Sur	0.01	0.01	-0.03	-0.03
	US	5y5y	-0.16	-0.18	0.14	0.15
		Sur	-0.17	-0.18	0.09	0.10
HO1	UK	5y5y	-0.36	-0.23	-0.20	-0.05
		Sur	0.00	0.00	0.03	0.03
	US	5y5y	-0.21	-0.21	0.16	0.16
		Sur	0.04	0.05	0.14	0.15
NG1	UK	5y5y	0.00	0.00	-0.04	-0.04
		Sur	-0.22	-0.24	-0.06	-0.07
	US	5y5y	0.00	0.00	0.00	0.00
		Sur	0.03	0.03	-0.12	-0.13
QS1	UK	5y5y	-0.20	-0.06	-0.07	0.08
		Sur	0.01	0.01	0.01	0.01
	US	5y5y	-0.12	-0.13	0.03	0.03
		Sur	-0.11	-0.11	0.02	0.02
XB1	UK	5y5y	-0.08	-0.12	0.12	0.11
		Sur	-0.02	-0.02	0.03	0.03
	US	5y5y	-0.26	-0.27	-0.11	-0.12
		Sur	-0.37	-0.38	0.16	0.15

Table 16: Sharpe difference between full-sample in-sample and out-of-sample methodologies

Then combining the returns into one portfolio leads to the following returns which are quite strong. The time series are equal weighting across all securities and signals. The signal returns are volatility targeted at 10% annualized portfolio, while the portfolio is not volatility targeted. The expanding out-of-sample portfolio performs the best which is likely due to some temporal model shift that it captures in the early periods.

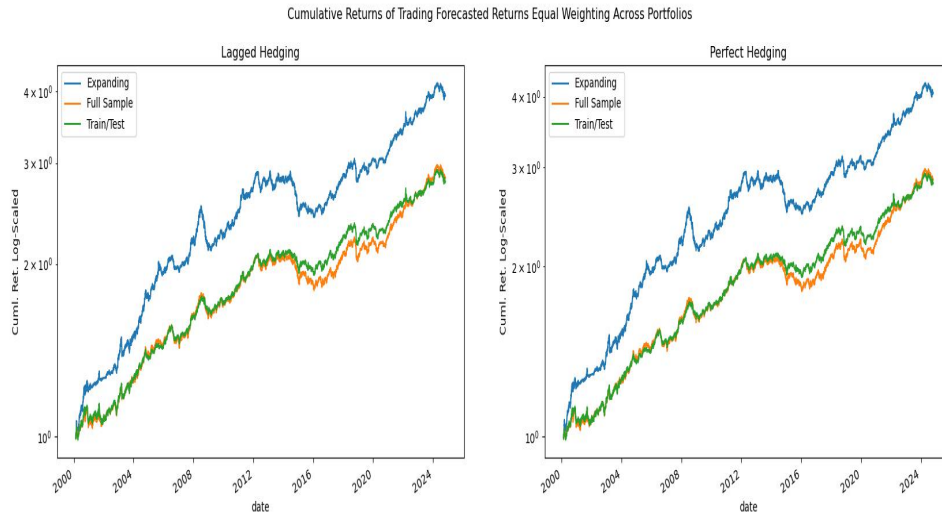


Figure 7: Performance of trading OLS forecasted, equal weighting, 10% annualized volatility

The expanding out-of-sample out-performance is evident when taking the returns difference between expanding returns and the two other portfolios. Most of the outperformance occurs during the sample period 2000-2018.

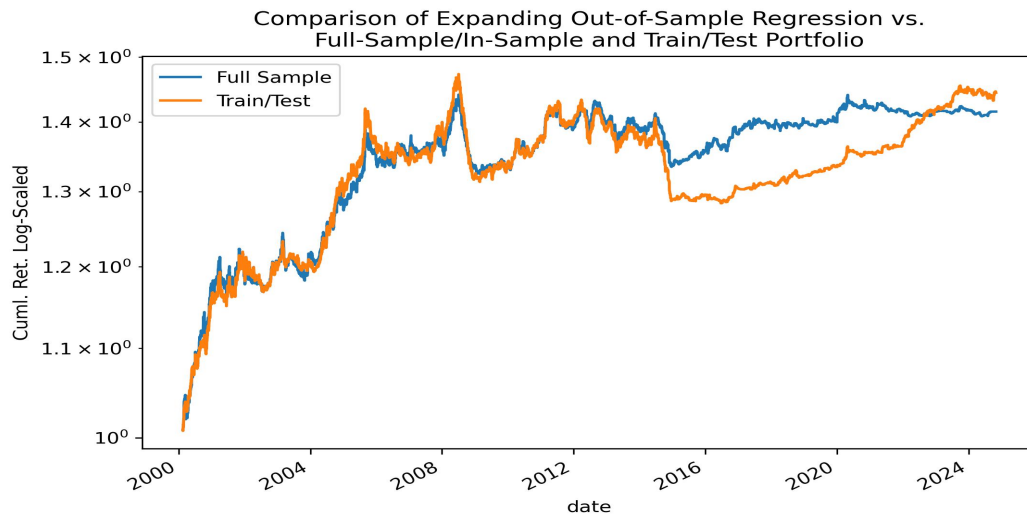


Figure 8: Comparative performance between Expanding Out-of-Sample of OLS forecasted values and other regressions, equal weighting, 10% annualized volatility (hedged contemporaneously)

The consistency of the portfolios controlling sample size leads to reasonably strong sharpes across the board. Below is a table of the sharpes of the portfolios.

Vol. Target	Lagged	Perfect
Expanding	0.96	1.01
Full Sample	0.77	0.81
Train/Test	0.85	0.88

Table 17: Sharpes OLS forecasted portfolios, equal weighting, 10% annualized volatility

5.3 Optimized Signal

Now that the signals have been created, tested within OLS, backtested, and the results have been verified and optimized signal approach can be used. This optimization is done by deciling values calculating the sharpe be decile bins. Then grouping the upper and lower 2 decile bins and taking a position of the corresponding sharpe if both sharpes within the same upper and lower group are the same. If that condition isn't met, then the signal isn't traded and neither are the remaining deciles.

This approach is applied to the residuals of the OLS and the z-score. They perform much better for the z-score as opposed to OLS residuals. The residual performance is also much more cross-correlated since the models are capturing the relatively the same factor.

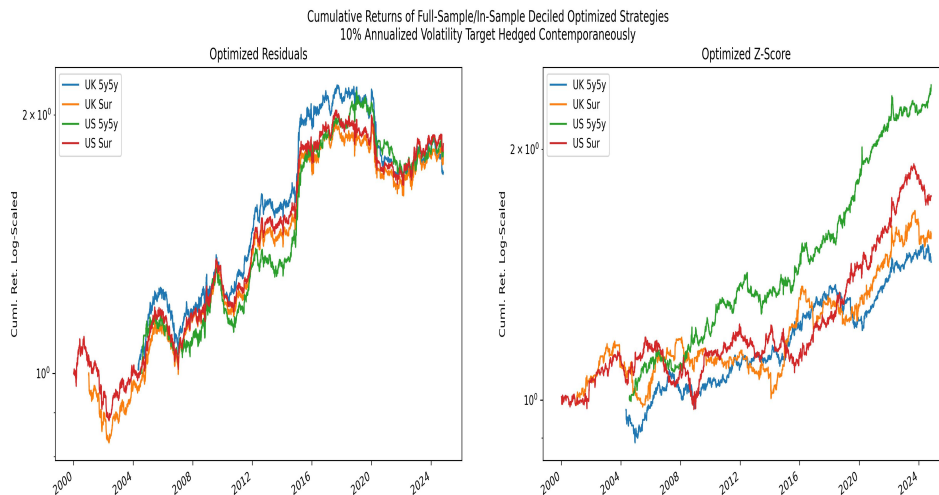


Figure 9: Deciled Optimized Strategies, 10% Annualized volatility, equal weighting

The signals returns are volatility targeted, accounting for the days when no position is taken. The resulting sharpes of the portfolios are substantial. The table below is the sharpes of the portfolio controlling for days when there isn't a position. Raw sharpe accounts for the optimized returns themselves, while adjusted accounts for days of no

position. At times the comparison can lead to quite different sharpes but since there are so many signals being applied across multiple futures the difference is quite small.

Country	Sharpe Signal	Optimized Residuals		Optimized Z-Score	
		Adjusted	Raw	Adjusted	Raw
UK	5y5y	0.51	0.44	0.69	0.44
	Sur	0.48	0.45	0.62	0.43
US	5y5y	0.54	0.48	1.42	0.90
	Sur	0.49	0.47	0.71	0.50

Table 18: Comparing adjusted and raw sharpes of decile optimized signals Full Sample/In-sample

The next step is to test the deciles out-of-sample. Expanding deciles is quite difficult to compute⁸ therefore training and testing periods will be picked using 2018. The deciling returns method leads to unique returns streams when changing the sample size. Normally the dispersion of results would be questionable, but the prior OLS factor analysis had shown that the signals are statistically significant. Below is a plot of the Full-Sample/In-Sample and Train/Test for the decile optimization.

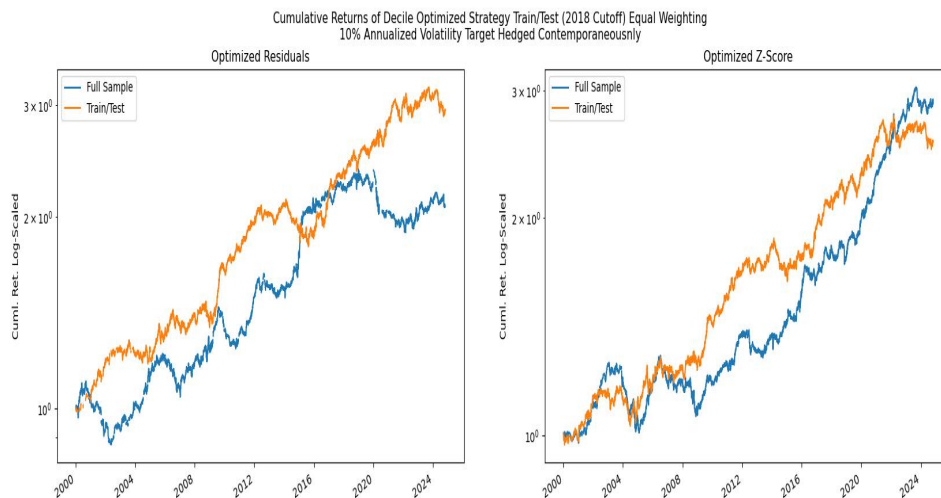


Figure 10: Comparing Full-Sample and Train/Test Sample Deciled portfolios, 10% Annualized volatility, equal weighting

⁸This is a future task for the codebsae

The resulting portfolios have reasonable good sharpes which can be seen below.

Sharpe Name	Full Sample		Train/Test	
	Adj	Raw	Adj	Raw
Avg	0.66	0.66	0.77	0.77
Optimized Residuals	0.54	0.47	0.78	0.78
Optimized Z-Score	0.82	0.75	0.70	0.70

Table 19: Sharpes of Full Sample and Train/Test Portfolios (adjusted and raw), 10% Annualized Volatility Target (Hedged Contemporaneously)

The lagged volatility targeting does lead to decreased performance, much greater in the prior examples. This greater difference likely stems from the volatility structure is a continuous variable of a non-continuous optimized return. Below is the table.

Sample Group	Hedging Signal	Lagged	Perfect
Full Sample	Avg	0.54	0.66
	Optimized Residuals	0.44	0.54
	Optimized Z-Score	0.75	0.82
Train/Test	Avg	0.66	0.77
	Optimized Residuals	0.67	0.78
	Optimized Z-Score	0.63	0.70

Table 20: Comparing Lagged vs. Contemporaneously Volatility Targeted Portfolios

It is not completely clear whether the optimized signals should be added into the OLS forecasted backtest. The optimized residual performance isn't vastly superior to the OLS forecasted model yet the cross-correlation⁹ to the forecasted model is quite low. Therefore the resulting portfolio doesn't perform drastically better.

Sample	Strategy Target	Avg	Forecasted	Optimized
Full Sample	Lagged	0.71	0.70	0.49
	Perfect	0.82	0.75	0.60
Train/Test	Lagged	0.84	0.84	0.66
	Perfect	0.93	0.87	0.78

6 Cross-Sectional Models

The prior models focused on building signals to apply to securities individually. A model can be built cross-sectionally. Regression residuals can be ranked and built into Long and Short legs for a market neutral portfolio. In commodities cross-sectional long short

⁹About ~30%, see in GitHub Repo

portfolios are quite difficult to manage since they can naturally lead to large net exposures.¹⁰ The use of energy futures provides the best hedge to managing oil exposures although some energy-related hedges can't be perfectly hedged.

The portfolio is tested with daily rebalancing and using two cross-sectional groups, due to the limited nature of commodity futures. The returns are volatility-targeted beforehand and equal weight is taken across the legs. Equal weighting amongst the legs is used. Since the returns are volatility targeted the portfolio is close to equal volatility contribution naturally. The historical performance is quite strong and below is a figure for the returns.

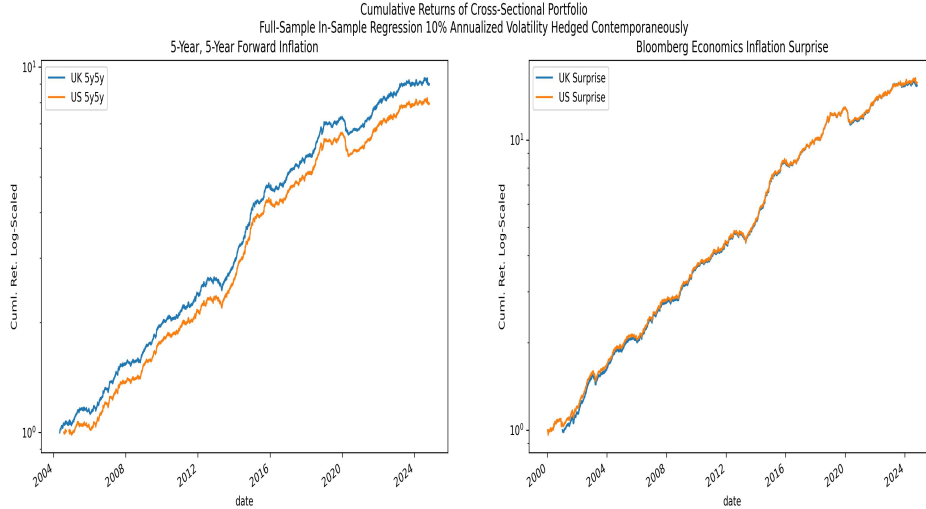


Figure 11: Long-Short Performance of OLS Residuals, In-Sample, security-level volatility targeting at 10%

The resulting portfolios are all highly correlated with each other, which implies that they are harvesting the same returns premium, thus strengthening the relationship. The cross-correlation is roughly 99%¹¹, and below is a table of the sharpes which are separated by long and short leg as well.

Country	Signal Name	Long Leg	Short Leg	Net
UK	5y5y	0.74	-0.52	1.94
	Surprise	0.82	-0.53	2.06
US	5y5y	0.68	-0.54	1.88
	Surprise	0.88	-0.42	1.97

Table 21: Long-Short Performance of OLS Residuals

To expand on the results and controlling regression sample size and volatility target leads to reasonably the same results.

¹⁰For example building a long short commodity portfolio may result in being long energy futures and short agriculture futures, which leads to un-hedged agriculture and energy exposures. In equities it can occur as well, but shorting stocks naturally hedge β which is less feasible possible in commodities

¹¹see Repo for Matrices

Regression	Signal	Target Leg Country	Lagged			Perfect		
			Long Leg	Short Leg	Net	Long Leg	Short Leg	Net
Expanding	5y5y	UK	0.75	-0.50	1.89	0.73	-0.51	1.90
		US	0.69	-0.51	1.82	0.67	-0.53	1.83
	Surprise	UK	0.87	-0.52	2.09	0.83	-0.54	2.08
		US	0.89	-0.43	1.97	0.86	-0.44	1.96
Full Sample	5y5y	UK	0.76	-0.51	1.92	0.74	-0.52	1.94
		US	0.70	-0.53	1.86	0.68	-0.54	1.88
	Surprise	UK	0.86	-0.52	2.07	0.82	-0.53	2.06
		US	0.92	-0.40	1.98	0.88	-0.42	1.97
Train/Test	5y5y	UK	0.76	-0.50	1.90	0.74	-0.51	1.92
		US	0.71	-0.53	1.86	0.68	-0.55	1.88
	Surprise	UK	0.85	-0.51	2.06	0.82	-0.52	2.05
		US	0.92	-0.40	1.98	0.88	-0.42	1.96

Table 22: Long-Short Performance of OLS Residuals, In-Sample, security-level volatility targeting at 10%

The sharpe lost from in sample to out-of-sample is quite small as well. The table below examines the relationship.

Country	Signal	Target	Expanding	Train/Test
UK	5y5y	Lagged	-0.04	-0.02
		Perfect	-0.04	-0.02
	Surprise	Lagged	0.02	-0.01
		Perfect	0.03	-0.01
US	5y5y	Lagged	-0.04	0.00
		Perfect	-0.05	-0.00
	Surprise	Lagged	-0.01	-0.01
		Perfect	-0.01	-0.01

Table 23: Sharpe performance degradation from going in-sample to out-of-sample

The loss regarding volatility targeting is also quite small as well. Ultimately there isn't much net volatility exposure within the returns due to the position neutrality.

Country	Signal	Expanding	Full Sample	Train/Test
UK	5y5y	0.01	0.01	0.01
	Surprise	-0.01	-0.01	-0.01
US	5y5y	0.01	0.02	0.02
	Surprise	-0.01	-0.02	-0.02

Table 24: Sharpe performance comparison between perfectly targeted returns and lagged targeting

References

- [1] David Andolfatto and Mahdi Ebsim. Oil prices and expected inflation. FRED Blog, Federal Reserve Bank of St. Louis, April 2020.